

Steps to Create a New Instance in Oracle OCI Console

1. **Log in to Oracle OCI:**
 - Navigate to the Oracle OCI console by visiting the URL <https://www.oracle.com/in/cloud/>.
 - Log in using your credentials to access the Oracle Cloud Infrastructure dashboard.
2. **Navigate to the Instance Tab:**
 - Once logged in, you'll be directed to the OCI home page.
 - From the navigation menu, select **Compute** and then click on **Instances**. This will take you to the page where you can manage existing instances or create a new one.
3. **Create a New Instance:**
 - Click on the **Create Instance** button to initiate the process.
 - Provide a **friendly name** for the new instance. This is the label used to identify the instance easily. The name should be descriptive, especially if managing multiple instances.
4. **Compartment Settings:**
 - In the compartment section, leave the compartment set to **Root** (unless you are organising resources into specific compartments).
 - This ensures that the instance is created under the main compartment unless another compartment structure is required.
5. **Select Image and Shape:**
 - Choose an **image** for the instance, which determines the operating system. In this case, select **Ubuntu with Canonical Ubuntu 22.04** from the list of available images.
 - Under the **Shape** section, choose between **AMD** or **Intel** processors depending on your performance needs.
 - This setting impacts the underlying hardware that powers the instance.
6. **Adjust CPU and RAM:**
 - Configure the number of **CPU cores** and the amount of **RAM** based on your application or workload needs.
 - Use the sliders or input fields to adjust the number of vCPUs and the amount of memory (RAM). Ensure these values align with the resource requirements of the services you plan to run.
7. **Enable Burstable Option:**
 - If required, select the **Burstable Instance Option**. This allows the instance to handle traffic spikes or increased load temporarily by utilizing extra resources (bursting).
 - This feature is helpful when workloads are variable and may need to temporarily exceed allocated resources.
8. **Network Settings:**
 - Under **Network settings**, select the appropriate **Virtual Cloud Network (VCN)** name from the available options.
 - Choose the **Subnet** where the instance will be placed. This ensures that your instance is part of the correct network and can communicate with other resources in the same network.
 - Decide whether to assign a **Private IP** or **Public IP**. If the instance will be accessed from outside the cloud environment (such as over the internet), a public IP is needed. Otherwise, a private IP is sufficient for internal cloud communication.
9. **SSH Key Setup:**
 - In the **SSH Keys** section, either **upload your existing SSH key pair** or let the system generate a new pair for you.
 - **Save both the private and public keys** on your local machine, as you'll use them to securely connect to the instance once it's running.
10. **Configure Boot Volume:**
 - Under the **Boot Volume** section, specify the required **disk size** for the instance.
 - Choose the storage size based on the application's needs, ensuring enough space for the operating system, applications, and any data the instance will store.
11. **Create the Instance:**

- After configuring all the settings, click on the **Create** button.
- The instance creation process will begin, and within a few minutes, the instance will be provisioned and **rebooted** to become ready for use.
- You'll see the instance's status change from **Provisioning** to **Running** when it is ready.

12. VM Connection:

- Once the instance is running, check its **connection** by accessing it via SSH using the **private key** you saved earlier.
- Use an SSH client like PuTTY (on Windows) or the terminal (on Linux/macOS) to connect to the instance with the provided IP address (either private or public depending on your choice during setup).

```
ssh -i path_to_private_key ubuntu@instance_public_IP
```

- Ensure that any required **firewall settings** and **security groups** are configured to allow access via the correct ports, such as SSH (port 22) for your connection.